Rolle's Theorem: Let f be continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b). If $f(a)=0$ and $f(b)=0$, then there is at least one point c in the interval (a,b) such that $f'(c)=0$.

Example: Find the x-intercepts of the function $f(x) = x^2-5x+4$ and confirm that $f'(c)=0$ at some point between the ~~inter~~ intercepts.

Solution: To find out the x-intercepts we set

$$f(x) = 0.$$

$$\Rightarrow x^2-5x+4=0$$

$$\Rightarrow x^2-4x-x+4=0.$$

$$\Rightarrow x = 1, 4$$

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Now we set

$$f'(x) = 0.$$

$$\Rightarrow 2x - 5 = 0$$

$$\Rightarrow x = \frac{5}{2}$$

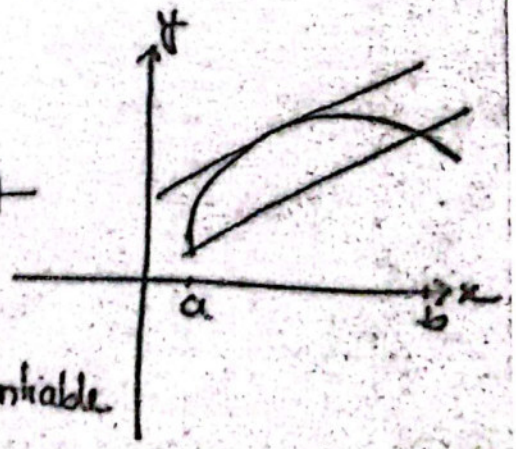
Therefore, $f'(x) = 0$ at the point $x = \frac{5}{2}$

Ans

Mean Value Theorem (MVT) :- Rolle's theorem is a special case of a more general result called Mean Value Theorem.

Let f be continuous on the closed interval $[a,b]$ and differentiable on the open interval (a,b), then there is at least one point c in (a,b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



Geometrically this theorem states that between any two points $(a, f(a))$ and $(b, f(b))$ on the graph of a differentiable function f, there is at least one place where the tangent line to the graph is parallel to the second line.

Example:- Show that the function $f(x) = \frac{1}{4}x^3 + 1$ satisfies the hypotheses of the Mean Value Theorem over the interval $[0, 2]$ and find all values of c in the interval $(0, 2)$.

Solution:- Given that

$$f(x) = \frac{1}{4}x^3 + 1.$$

$$f(0) = 0 + 1 = 1$$

$$f(2) = \frac{1}{4} \times 2^3 + 1 = \frac{5}{4}\ 3$$

We set,

$$f'(x) = \frac{3}{4}x^2$$

and $\frac{f(b) - f(a)}{b - a} = \frac{f(2) - f(0)}{2 - 0}$

$$= \frac{3 - 1}{2 - 0}$$

$$= \frac{2}{2} = 1.$$

According the hypotheses of the Mean Value Theorem

$f'(x) = 1$

Therefore $x = \frac{2}{\sqrt{3}}$.

* Squeeze theorem: $\boxed{\text{Thm: 1.6.4, H.A, page-123}}$

Let $f(x)$, $g(x)$, $h(x)$ be functions satisfying

$$g(x) \leq f(x) \leq h(x).$$

for all x in ~~the~~ some open interval containing the number c, with the possible exception that the inequalities need not hold at c.

If $g(x)$ and $h(x)$ have the same limit as x approaches c,

say $\lim_{x \to c} g(x) = \lim_{x \to c} h(x) = L$

then $f(x)$ also has this limit as x approaches to c, that is

$$\lim_{x \to c} f(x) = L.$$

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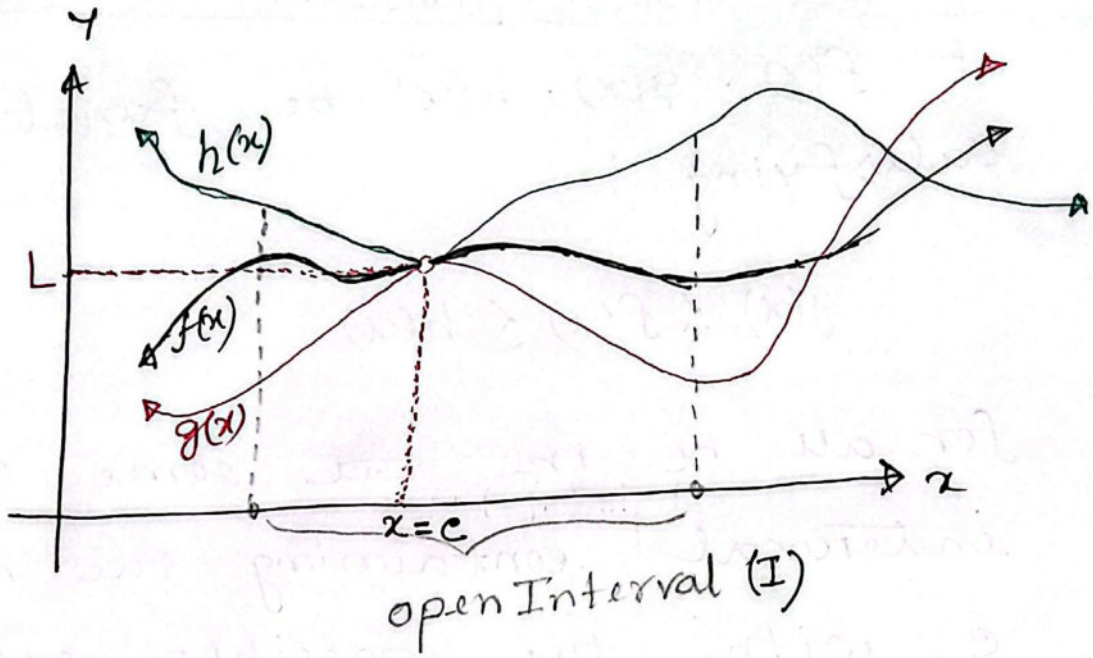


$$g(x) \le f(x) \le h(x) \ , \ \forall x \in I .$$

$$\Rightarrow \lim_{x \to c} g(x) \le \lim_{x \to c} f(x) \le \lim_{x \to c} h(x)$$

$$\Rightarrow L \le \lim_{x \to c} f(x) \le L$$

$$\Rightarrow \lim_{x \to c} f(x) = L .$$

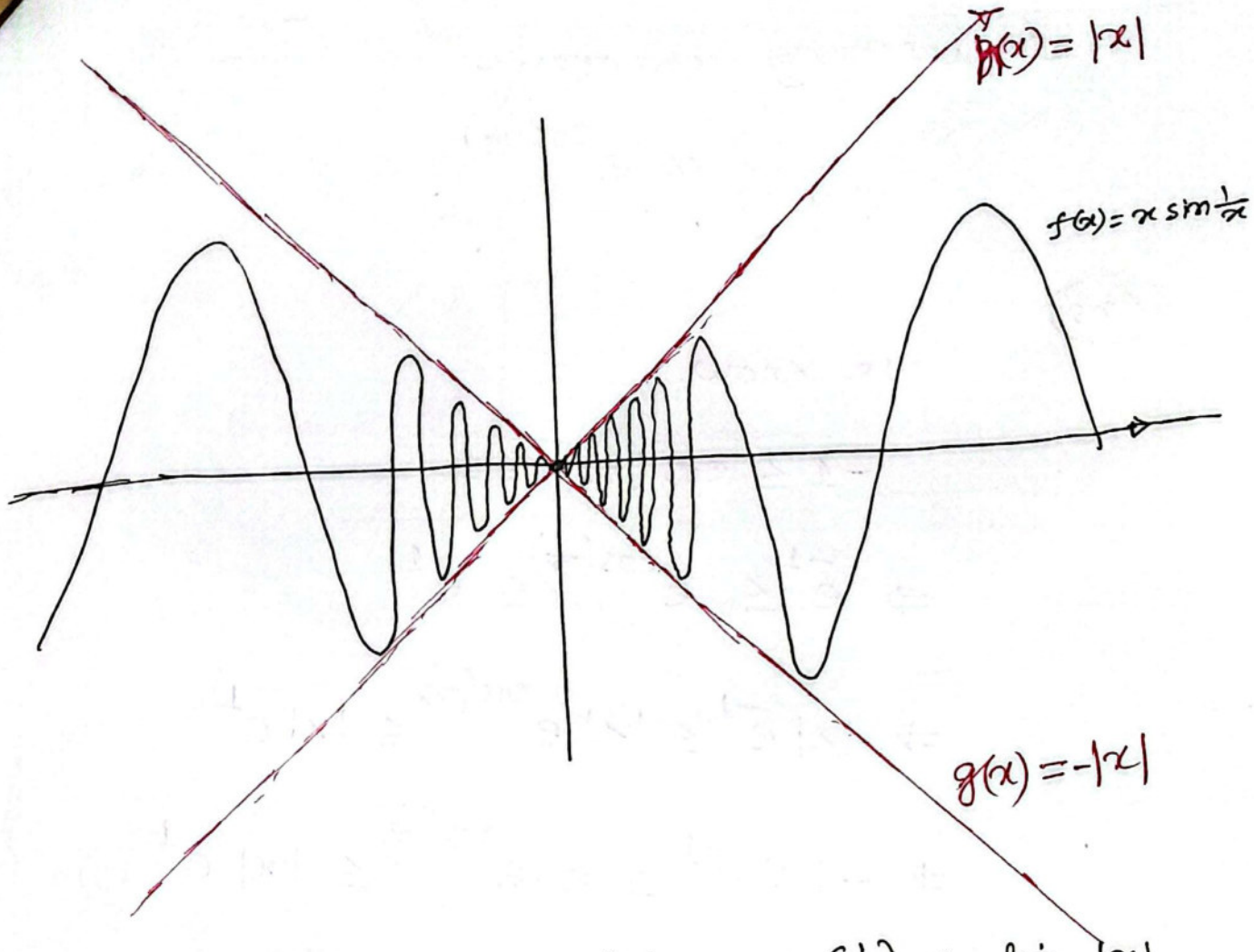
★ Find $\lim_{x \to 0} x \cdot \sin\left(\frac{1}{x}\right)$ using squeezing theorem.

Ans: we know,

$$-1 \le \sin\left(\frac{1}{x}\right) \le 1$$

$$\Rightarrow -1 \times |x| \le |x| \sin\left(\frac{1}{x}\right) \le 1 \times |x|$$

$$\Rightarrow -|x| \le x \sin\left(\frac{1}{x}\right) \le |x|$$



$$\Rightarrow \lim_{x \to 0} -|x| \leq \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) \leq \lim_{x \to 0} |x|$$

$$\Rightarrow 0 \leq \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) \leq 0$$

$$\therefore \lim_{x \to 0} x \sin\left(\frac{1}{x}\right) = 0$$

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* Evaluate the limit using squeezing theorem

$$\lim_{x \to 0} x e^{\cos\left(\frac{1}{x}\right)}.$$

Ans:

we know,

$$-1 \le \cos\left(\frac{1}{x}\right) \le 1$$

$$\Rightarrow e^{-1} \le e^{\cos\left(\frac{1}{x}\right)} \le e^{1}$$

$$\Rightarrow -|x| e^{-1} \le |x| e^{\cos\left(\frac{1}{x}\right)} \le |x| e^{1}$$

$$\Rightarrow -|x| e^{-1} \le x e^{\cos\left(\frac{1}{x}\right)} \le |x| e^{1}$$

$$\Rightarrow \lim_{x \to 0} -|x| e^{-1} \le \lim_{x \to 0} x e^{\cos\left(\frac{1}{x}\right)} \le \lim_{x \to 0} |x| e^{1}$$

$$\Rightarrow 0 \le \lim_{x \to 0} x e^{\cos\left(\frac{1}{x}\right)} \le 0$$

$$\therefore \lim_{x \to 0} x e^{\cos\left(\frac{1}{x}\right)} = 0.$$

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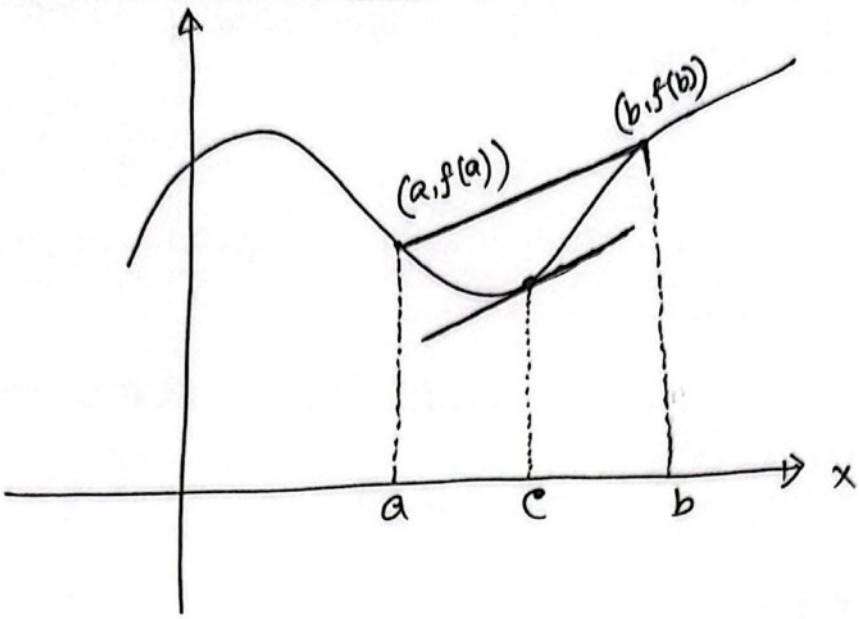
* Mean Value Theorem



$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Ex. $f(x) = \frac{1}{3}x^3 + 1$

$$= 1 + x + x^2 + x^3 + \cdots$$

Ans.

Optimization Problem

Example:- find the dimension of the rectangle with maximum area that can be inscribed in a circle of radius 10 cm.

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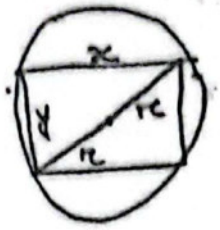
Solution:- Let



r : Radius of the circle

x : Length of the rectangle

y : Width of the rectangle

A : Area of the rectangle

Here

$$A = xy \quad \cdots \quad (1)$$

From the figure,

$$x^2 + y^2 = (2r)^2$$

$$\Rightarrow x^2 + y^2 = 4r^2$$

$$\Rightarrow y^2 = 4r^2 - x^2$$

$$\Rightarrow y = \sqrt{4r^2 - x^2}$$

We need to maximize A so that it can be inscribed in the circle of radius 10cm. Also we know that $0 \leq x, y \leq 20$. Differentiating A with respect to x,

$$\frac{dA}{dx} = \sqrt{400-x^2} - \frac{x^2}{\sqrt{400-x^2}}$$

Setting $\frac{dA}{dx} = 0$.

$$\Rightarrow \sqrt{400-x^2} - \frac{x^2}{\sqrt{400-x^2}} = 0.$$

$$\Rightarrow \frac{400-2x^2}{\sqrt{400-x^2}} = 0.$$

$$\Rightarrow 2x^2 = 400$$

$$\Rightarrow x = 10\sqrt{2}.$$

$$\frac{d^2A}{dx^2} = \frac{\sqrt{400-x^2}\,(-4x) - (400-2x^2)\,\frac{(-2x)}{2\sqrt{400-x^2}}}{400-x^2}$$

$$= \frac{-4x\sqrt{400-x^2} + x\,\frac{400-2x^2}{\sqrt{400-x^2}}}{400-x^2}.$$

< 0 at $x = 10\sqrt{2}$,

Therefore, the dimensions of the rectangle are $10\sqrt{2}$ cm and $10\sqrt{2}$ cm.

Example:- A liquid form of antibiotic manufactured by a pharmaceutical firm is sold in bulk at a price of \$200 per unit. If the total production cost (in dollars) for x units is,

$$C(x) = 500000 + 80x + 0.03x^2$$

and if the production capacity of the firm is at most 30,000 units in a specified time, how many units of antibiotic must be manufactured and sold in that time to maximize the profit?

Solution:- Let the amount need to be produced be x units.

Total revenue, $R(x) = 200x$.

Profit = Total revenue − Total cost

$\Rightarrow P(x) = R(x) - C(x)$

$\Rightarrow P(x) = 200x - 500000 - 80x - 0.03x^2$.

Setting $\frac{dP}{dx} = 0$.

$\Rightarrow 120 - 0.06x = 0$

$\Rightarrow x = 20,000$.

Here,

$$\frac{d^2P}{dx^2} = -0.06 < 0.$$

Since the second derivative is greater than zero all values of x, ~~there~~ the profit is maximized at $x = 20{,}000$ units.

Ans.

~~Example:~~

Example:- Find a point on the curve $y = x^2$ that is closest to the point $(18, 0)$.

Solution:- Let any arbitrary point on the curve be (x, y). The distance L between $(18,0)$ and (x,y) is,

$$L = \sqrt{(x-18)^2 + (y-0)^2} = \sqrt{(x-18)^2 + y^2}.$$

Since (x, y) lies on the curve, x and y satisfy $y = x^2$; thus,

$$L = \sqrt{(x-18)^2 + x^4}.$$

The distance L and the square of the distance L^2 are minimized at the same value. Thus the minimum value of $S = L^2 = (x-18)^2 + x^4$ occurs at the same x-value.

$$\frac{dS}{dx} = 2(x-18) + 4x^3 = 4x^3 + 2x - 36$$

So the critical points satisfy

$$4x^3 + 2x - 36 = 0$$

$$\Rightarrow 2x^3 + x - 18 = 0$$

$\Rightarrow$ $x = 2$ is the only real solution

To determine the nature of this critical point we will use the second derivative test.

$$\frac{d^2S}{dx^2} = 12x^2 + 2, \quad \text{so} \quad \left.\frac{d^2S}{dx^2}\right|_{x=2} = 50 > 0.$$

which show that a relative minimum occurs at $x = 2$.

Thus, the point of the curve $y = x^2$ closest to $(18, 0)$ is

$$(x, y) = (x, x^2) = (2, 4).$$

Ans.

Optimization Problem

① A garden is said to be laid out in a rectangular area and protected by a chicken wire fence. What is the largest possible area of the garden if only 100 running feet of chicken wire is available for the fence?

② Find a point on the curve $y = x^2$ that is closest to the point $(18, 0)$.

③ A chemical manufacturer sells sulfuric acid in bulk at a price of \$ 100 per unit. If the daily total production cost in dollars for x units is

$$C(x) = 100\,000 + 60x + 0.0025x^2$$

and if the daily production capacity is at most 7000 units, how many units of sulfuric acid must be manufactured and sold daily to maximize the profit?

④ A fertilizer producer finds that it can sell its product at a price of $p = 300 - 0.1x$ dollars per unit when it produces x units of fertilizer. The total production cost (in dollars) for x units is

$$C(x) = 15000 + 125x + 0.025x^2$$

If the production capacity of the firm is at most 1000 units of fertilizer in a specified time, how many units must be manufactured and sold in that time to maximize the profit?

⑤ A firm determines that x units of its product can be sold daily at p dollars per unit where

$$x = 1000 - p.$$

The cost of producing x units per day is

$$C(x) = 3000 + 20x.$$

a) Find the revenue function $R(x)$.

b) Find the profit function $P(x)$.

c) Assuming that the production capacity is at most 500 units per day, determine how many units the company must produce and sell each day to maximize the profit.

d) Find the maximum profit.

e) What price per unit must be charged to obtain the maximum profit?

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